
UNIT 12 NMR SPECTROSCOPY

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12.1 INTRODUCTION

In this course, you have so far learnt about various spectroscopic methods involving quantised electronic, vibrational and rotational energy states of molecules and the electronic states of atoms. In these methods, molecules and atoms are subjected to electromagnetic radiation of appropriate wavelength and resultant absorption, emission or scattering of radiation is measured. Now you will learn about nuclear magnetic resonance (NMR) spectroscopy where transitions between different nuclear spin states are involved. You would recall from Unit 1 that the quantised nuclear spin states come into existence when the sample is placed in an external magnetic field.

In this unit, you would learn about the theory behind the phenomenon of NMR and the types of nuclei that exhibit it. Thereafter you will learn about the characteristics like chemical shift, spin-spin coupling etc. of NMR spectra, their origin, the factors affecting them and the structural information carried by them. This will be followed by the instrumentation and the experimental set up required to obtain the NMR spectra. In the end we will discuss about some applications of NMR spectroscopy. You must have heard of MRI which is a modern medical diagnostic tool. MRI is also based on the phenomenon of nuclear magnetic resonance and finds widespread applications in the field of medicine.

Objectives

After studying this unit, you should be able to:

- state the type of nuclei that show the phenomenon of NMR,
- explain the basic principle of NMR,
- draw a schematic diagram of NMR spectrometer,
- explain the basic principle and advantages of Fourier transform NMR,
- describe the relaxation phenomenon and its mechanism,
- define and explain chemical shift and state the factors affecting it,
- explain the process of spin-spin splitting,
- correlate the NMR spectrum of simple molecules with their structure, and
- describe the applications of NMR spectroscopy in structure elucidation.

12.2 THEORY OF NMR SPECTROSCOPY

In nuclear magnetic resonance (NMR) spectroscopy, the magnetic properties of certain nuclei are exploited to seek structural information of the molecule. In order to understand the phenomenon of NMR we need to know about the nuclei that exhibit this phenomenon, their magnetic properties that make it possible and the meaning of resonance. Let us learn about these in the following subsections. We begin with the types of nuclei.

12.2.1 Types of Nuclei

You know from your earlier classes that the atomic nucleus is positively charged and some of the nuclei are associated with a fundamental property called **spin**. A nonspinning and spinning nucleus may be represented as shown in Fig. 12.1. The angular momentum of the spinning nucleus is defined in terms of **spin angular momentum quantum number, I** . The magnitude of spin angular momentum $|\mathbf{I}|$ of the spinning nucleus is related to the spin angular momentum quantum number, as follows:

$$|\mathbf{I}| = \sqrt{I(I+1)} \hbar \quad \dots(12.1)$$

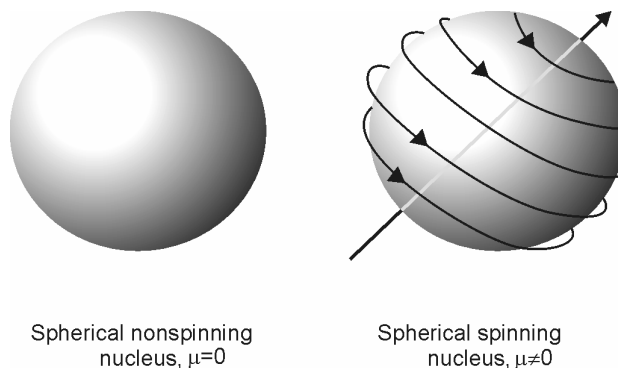


Fig. 12.1: Illustration of non spinning ($I = 0$) and spinning ($I = 1/2$) nucleus

The spin angular momentum of a nucleus is vector sum of the spin angular momenta of the component particles of the nucleus, namely the neutrons and protons. The exact

way in which the neutrons and protons are vectorially coupled can be understood from nuclear shell model. However, at this stage it is sufficient to know about some generalisation for the spin quantum number of different nuclei:

- i) Nuclei having even number of protons and neutrons have $I = 0$. The examples of such nuclei are ^4He , ^{12}C and ^{16}O .
- ii) Nuclei having odd number of protons and neutrons have integral value of I . For example, ^2H and ^{14}N have $I = 1$.
- iii) Nuclei having odd value for the sum of protons and neutrons have half integral value of I ; for example, ^1H and ^{15}N have $I = \frac{1}{2}$ and ^{17}O has $I = \frac{5}{2}$.

The nuclei with $I = \frac{1}{2}$ are important for NMR; proton (^1H) being probably the most exploited nuclei in NMR.

12.2.2 Magnetic Moment

Any nucleus with a spin angular momentum quantum number, $I \neq 0$, corresponds to a spinning positive charge and you know that any spinning charge will generate a magnetic moment (μ). The magnetic moment, μ , of a spinning nucleus is proportional to its spin angular momentum (I) and is given by the following expression.

$$\mu = \frac{g_N e}{2m} I \quad \dots (12.2)$$

where, g_N is called the **nuclear g-factor** which is characteristic of the particular nucleus (for proton its value is 5.585), e is the charge on a proton and m is the mass of the proton. The magnetic moment vector is in the same direction as the angular momentum vector (Fig. 12.2).

In a nucleus with $I = \frac{1}{2}$ the charge distribution is assumed to be spherical. However, all nuclei having $I = 1, 3/2, 2$ and more are nonspherical and have quadrupole moment (eQ) which is essentially a measure of deviation (or distortion) from the spherical shape.

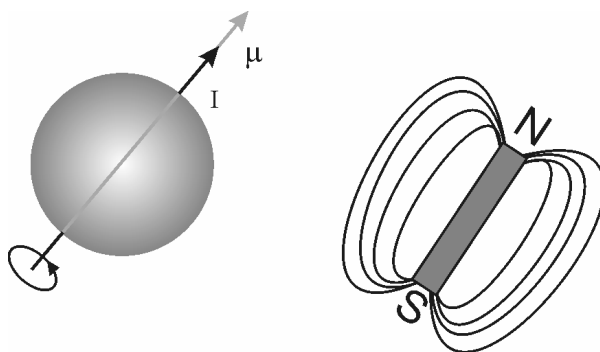


Fig. 12.2: Spin angular momentum and the magnetic moment having the same direction

We can get the relationship between the magnitude of magnetic moment and spin angular momentum quantum number by substituting the value of the magnitude of spin angular momentum in Eq. 12.2 from Eq. 12.1, as follows.

$$\begin{aligned} |\mu| &= \frac{g_N e}{2m_p} \sqrt{I(I+1)} \hbar \\ &= g_N \mu_N \sqrt{I(I+1)} \quad \dots (12.3) \end{aligned}$$

where $\mu_N = \frac{e \hbar}{2m_p}$ and is called the **nuclear Bohr magneton**.

SAQ 1

Which of the following nuclei will have magnetic moment and will be NMR active?

^2D , ^6Li , ^{11}B , ^{15}N , ^{18}O , ^{19}F , ^{23}Na , ^{24}Mg , ^{27}Al , ^{31}Si , ^{31}P , ^{32}S , ^{37}Cl , ^{39}K , ^{40}Ca .

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12.2.3 Quantisation

As mentioned above, all nuclei having nonzero spin angular momentum are associated with magnetic moment and hence may be assumed to be behaving as small bar magnets. However, you must remember that the nuclear spin magnet is a quantum particle unlike a small bar magnet and hence may take up only certain allowed orientations. A nucleus with spin quantum number, I , when placed in an external magnetic field, can assume $(2I + 1)$ orientations. Thus, a nucleus with $I = \frac{1}{2}$, can take just two $(2I + 1 = 2 \times \frac{1}{2} + 1 = 1 + 1 = 2)$ orientations. The z-component of spin angular momentum, I_z is quantised and its value in the direction of the applied magnetic field is given as follows.

$$I_z = m_I \hbar$$

where m_I is the quantum number for z-component of the spin angular momentum and it can take values of $-I, \dots, +I$. The possible orientations for $I = \frac{1}{2}$ are represented in Fig. 12.3.

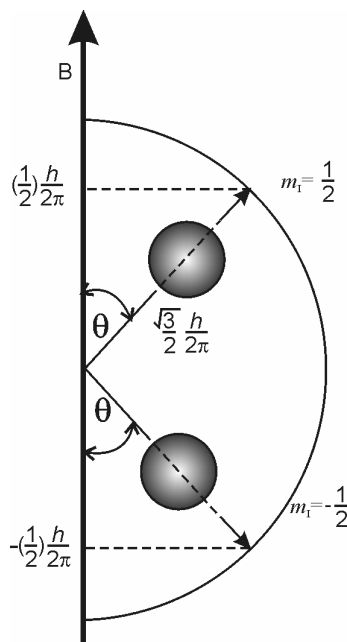


Fig. 12.3: Possible orientations of nuclear spin with $I = \frac{1}{2}$ in a magnetic field B_0

You can visualise the orientations of nuclei with spin $\frac{1}{2}$, as either aligned parallel or antiparallel (or opposite) to the external field as shown in Fig. 12.3. Any other orientation is not possible or permissible. Similarly, for $I = 1, \frac{3}{2}$ and 2 , the number of possible orientations will be 3, 4 and 5 respectively. The different orientations of the nuclear magnetic moment for the nuclei are associated with different energies. The energies being equal to $-m_I g_N \mu B_0$, where B_0 is the strength of the applied magnetic

field (applied in the z direction). The energy levels for $I = \frac{1}{2}$ nuclei are given in Fig.12.4.

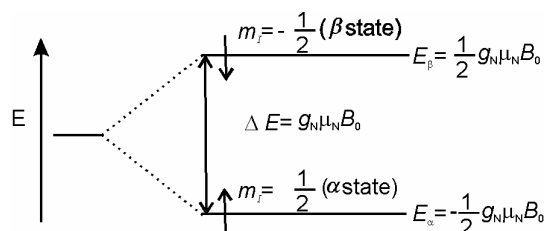


Fig. 12.4: Energy levels corresponding to $m_I = +\frac{1}{2}$ (lower level) and to $m_I = -\frac{1}{2}$ (upper level) with energy separation of ΔE

Traditionally, the energy level (or *state*, as it is commonly called) with $m_I = \frac{1}{2}$ is denoted as α and is sometimes described as ‘**spin up**’ and the state with $m_I = -\frac{1}{2}$ is denoted as β and is described as ‘**spin down**’. For the nuclei we are interested in, it is the α state which is the state with the lower energy. The difference in the energies, ΔE , of the two states is given below:

$$\Delta E = \left(\frac{1}{2} g_N \mu_N B_0 - \left(-\frac{1}{2} g_N \mu_N B_0 \right) \right) = g_N \mu_N B_0 \quad \dots (12.4)$$

This implies that the frequency of the radiation required for a transition from the lower level to the upper level would be given by:

$$\nu = \frac{g_N \mu_N \cdot B_0}{h} \quad \dots (12.5)$$

For the magnetic fields used in the NMR instruments, this frequency falls in the radiofrequency region of the spectrum. In other words, a suitable radiofrequency radiation can bring about the transition from the α to β spin state.

12.2.4 Population of Energy levels

When placed in an external magnetic field, all the spin magnetic moments in the sample, do not occupy the lowest available energy state. You know that the population of different energy levels is governed by **Boltzmann distribution law**. You would recall from Unit 1 that if n_1 and n_2 are number of spins in the lower and upper energy states respectively, their ratio at a given temperature, is given by

$$\frac{n_2 \text{ (upper)}}{n_1 \text{ (lower)}} = e^{(-\Delta E/kT)} \quad \dots (12.6)$$

where, ΔE is the energy difference between the two energy levels and k is the Boltzmann constant ($k = 1.380658 \times 10^{-23}$ J/K). As the energy difference between the two spin states is very small, the upper level will always be appreciably populated at all the temperatures above absolute zero (0 K). Let us calculate this ratio for protons in a magnetic field of 1.5 T at room temperature (300 K); ΔE is calculated using the Eq. 12.4.

$$\begin{aligned} \Delta E &= g_N \mu_N B_0 \\ &= 5.585 \times 5.05 \times 10^{-27} \text{ J T}^{-1} \times 1.5 \text{ T} \\ &= 4.23 \times 10^{-26} \text{ J} \end{aligned}$$

Since $\Delta E/kT$ is very small, $e^{(-\Delta E/kT)}$ may be approximated to $(1 - \Delta E/kT)$. Thus, the ratio of nuclear spin populations in two energy states will be:

$$= 1 - (4.23 \times 10^{-26} / 1.380658 \times 10^{-23} \times 300) = 1 - 1.02 \times 10^{-5} \\ = 0.9999898$$

Since transition from lower state to upper state involves absorption of radiation and that from the higher state to lower state the emission of radiation, no absorption can occur unless lower energy level has excess of protons. In the example given above, it is found that the lower energy level has about 2 excess spins for every 10^5 spins in the upper level. Though, it is a very small number but it is finite and hence absorption is observed. In the absence of this small excess population, no NMR can be observed.

SAQ 2

Calculate the ratio of number of nuclei in the upper energy state to the number in lower energy state of ^{13}C nucleus in a field of 2.3 T at 300 K. Given that $g_N = 1.405$ for ^{13}C and $\mu_N = 5.05 \times 10^{-27} \text{ J T}^{-1}$.

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You have learnt that when the spinning nuclei with $I = \frac{1}{2}$ are kept in magnetic field, their energies are quantised and they are allowed to take up any of the two permissible orientations. In addition to the quantisation, another very significant effect comes into play. This is called **Larmor precession**. It is essential to understand this phenomenon to understand the phenomenon of nuclear magnetic resonance. Let us learn about it.

12.2.5 Larmor Precession

In order to understand the phenomenon of Larmor precession, you need to recall the behaviour of a spinning top (or a gyroscope). You would recall that the spinning top executes two simultaneous motions. It spins on its axis as well as around its axis. The motion around the vertical axis arises due to the interaction of spin, i.e., gyroscopic motion with the earth's gravity acting vertically downward. This motion is called **precessional motion** and the spinning top is said to be precessing around the vertical axis of the earth's gravitational field as depicted in Fig. 12.5. You should also remember that only a spinning top undergoes precessional motion whereas static top does not do so.

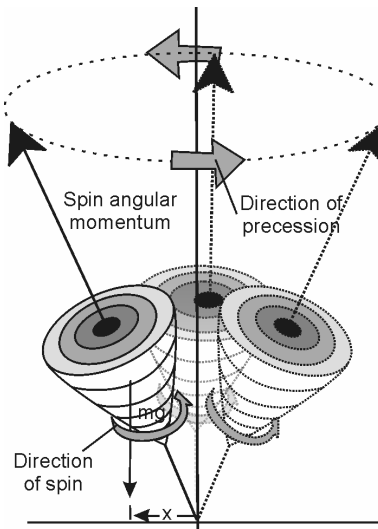


Fig. 12.5: Precessional motion of a spinning top due to earth's gravitational field

A spinning nucleus (or proton) when placed in an external magnetic field also exhibits precessional motion. A spinning nucleus under the presence of an external magnetic field can precess around the axis of an external magnetic field in two ways; it can either align with the field (low energy state) or it can oppose the field direction (high energy state) as typically illustrated in Fig.12.6 where B_0 is the external magnetic field. ΔE represents the energy of the transition between two orientations.

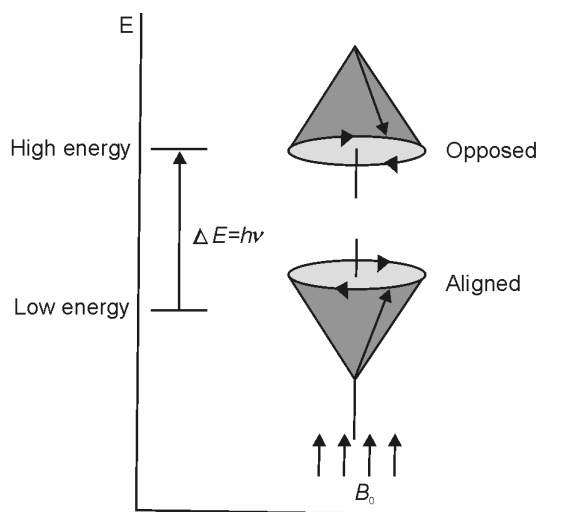


Fig. 12.6: Representation of precession of nucleus in two orientations; aligned to the external magnetic field and opposed to the magnetic field

The precessional frequency (ω) also called **Larmor frequency** is directly proportional to the strength of the external magnetic field (B_0). Thus

$$\omega \propto B_0 \quad \text{or} \quad \omega = \gamma B_0$$

where, γ is called the **magnetogyric** (or **gyromagnetic**) ratio. It is the ratio of the nuclear magnetic moment (μ) and the nuclear angular momentum (I). i.e., $\gamma = \mu/I$. The normal frequency, ν and the precessional frequency, ω are related as follows:

$$\omega = 2\pi\nu = \gamma B_0 \quad \dots (12.7)$$

$$\text{or } \nu = \frac{\gamma}{2\pi} B_0 = \frac{B_0}{2\pi} \cdot \frac{\mu}{I} \quad \dots (12.8)$$

After substituting values of I and μ , from Eqs. 12.1 and 12.3 respectively, we get the following expression.

$$\nu = \frac{g_N \mu_N B_0}{h} \quad \dots (12.5)$$

This is the same as Eq. 12.5 described in subsection 12.2.3. Therefore, the precessional frequency of the spinning nucleus is exactly equal to the frequency of electromagnetic radiation necessary to induce a transition from one nuclear spin state to another spin state. This fact provides a mechanism of causing the nuclear spin transition.

12.2.6 Mechanism of Resonance

Experimentally, there are two different ways to achieve resonance. To understand these, let us have a look at the Eq. 12.5 again. The equation shows that there are two variable parameters: (i) frequency (ν) and (ii) magnetic field (B_0); g_N , μ_N and h being constants. One obvious way to achieve resonance is to vary the frequency at a fixed

Miscellaneous Methods

The condition at which the frequency of the radiofrequency radiation matches with the Larmor precessional frequency is called the **resonance**.

magnetic field strength. This technique is called the **frequency sweep method**. In this method the sample is placed between the poles of a strong magnet which generates two energy levels and causes the nucleus to undergo precessional motion. The sample is then irradiated with a variable radiofrequency generated by an oscillator or transmitter. The resonance condition is achieved when the frequency of the radio frequency radiation matches with the Larmor precessional frequency. Under this condition called the **resonance**, the nuclei in the lower spin state absorb the radiation and go over to the higher energy state. This transition is commonly called as '**spin flip**'.

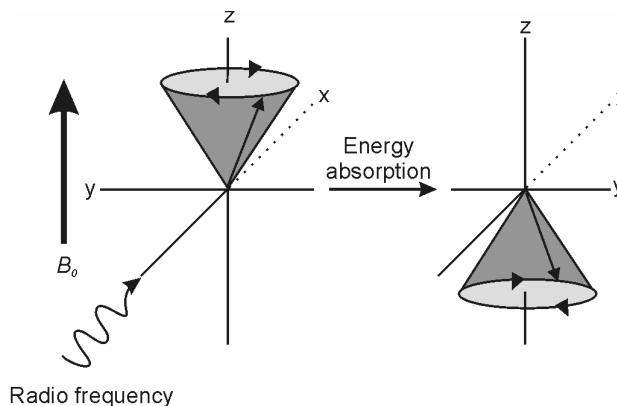


Fig. 12.7: Spin flip: Absorption of the radio frequency radiation causes the α spin to change (flip) to β spin

Alternatively, we can keep the oscillator frequency constant and vary the magnetic field strength. This technique is called the **field sweep method**. When the field strength is low the precessional frequency is lower than the applied radiofrequency and since the two frequencies do not match, no energy is absorbed. As we increase the field, the precessional frequency starts increasing and at a certain field strength it matches exactly with the applied radio frequency and the resonance condition is achieved and the radiation is absorbed. This causes the transition from α to β state. In the common instruments, both methods are used; however, the field-sweep method is preferred. It is because of relative ease of varying the field. In field sweep spectrometers the continuous variation of the magnetic field strength is achieved with the help of special coils present at the poles of the magnet.

In both the set ups the different protons (nuclei) are brought into resonance one by one by continuously varying either the field or the frequency. Therefore this is called **continuous wave (CW) spectrometry**. The CW method was the basis of all NMR instruments constructed up to about the end of the 1960s. Though CW is still used in some lower resolution instruments, most of the modern instruments use pulsed Fourier transform (FT) technique (discussed in Sec 12.3). However, for the interpretation of NMR spectra it does not matter whether the NMR spectrum is recorded by the CW or FT technique.

12.2.7 Relaxation Mechanisms

You have learnt in subsection 12.2.4 that the population of the energy levels is governed by Boltzmann distribution law. Once a radiation is absorbed, the population distribution gets disturbed. As the population difference between the two spin states is very small, the population of lower and higher energy levels would become equal after absorption of radiation for some time. It means that resonance signal has achieved saturation and no more energy can be absorbed. However, there are some competing phenomena that do not allow this stage to come. These are called **relaxation mechanisms**. In the process of relaxation the nuclear spins in the excited state relax

down to the lower state through radiationless processes. There are of two types of relaxations namely, **spin-lattice relaxation** and **spin-spin relaxation**. In order to maintain the population difference in the energy levels, the rate of relaxation from the excited state to the lower energy level must be greater than the rate at which radiofrequency is absorbed. Let us learn about these relaxation mechanisms.

Spin-lattice relaxation: This process involves the transfer of energy from the nuclear spin in the high energy state to the molecular lattice. The random motion of the adjacent nuclei in the lattice set up fluctuating magnetic fields at the nucleus leading to its interaction with the magnetic dipole of the excited nucleus. This results in the transfer of energy from the excited nucleus to the neighbouring atoms. This is also called **longitudinal relaxation**. Thus, a nucleus returns to its original low energy state from the excited state and excess of nuclei are maintained in the low energy state which is a necessary condition for the energy to be absorbed.

The term lattice refers to molecular framework of solid/liquid/gaseous sample and the solvent.

Spin-spin relaxation: This mechanism of relaxation involves mutual exchange of spins by two precessing nuclei in close proximity of each other. For example, if there are two nuclei then one of these can flip down and the other may flip up by mutual exchange of spins. This is also called **transverse relaxation**. It shortens the life time of an individual nucleus in the higher state but does not contribute to the maintenance of the required excess in the lower energy state. Due to the relaxation mechanisms, the excited states have small life times.

The lifetime of excited state is inversely related to the width of absorption line. As a result of inverse proportionality between the line width and lifetime of the excited state sharp resonance lines are observed for nuclei having long lived excited states and broad lines for short lived states. Thus, both spin-spin and spin-lattice relaxation processes contribute to the line width.

SAQ 3

Fill in the blanks spaces in the following with appropriate words.

- The ratio of the nuclear magnetic moment and the nuclear angular momentum is called the
- The continuous variation in the field or the frequency to achieve resonance during the NMR spectroscopy measurements gives rise to
- The process of transfer of energy from nuclear spin in the high energy state to the molecular lattice is called the whereas the exchange of spins during relaxation is referred to
- Relaxation contributes towards maintaining the excess population of the ground state.

12.2.8 Nuclei other than Protons

You would recall that only those nuclei show NMR that have nuclear spin angular momentum quantum number, $I \geq \frac{1}{2}$. More than 200 isotopes in principle can be studied by NMR. Of these, ^{13}C , ^{19}F and ^{31}P are most widely used besides ^1H . The characteristics of these nuclei are given in Table 12.1. In many organic compounds where proton magnetic resonance spectrum does not provide unambiguous information, carbon-13 NMR spectra have been found to be especially useful for structure elucidation. ^{13}C NMR spectra are intrinsically much less sensitive than proton NMR spectrum. The low signal strength results from the fact that the natural abundance of ^{13}C is only 1.11% and also its magnetic moment is $\frac{1}{4}$ th of that of the proton in a given magnetic field (Table 12.1).

Table 12.1: Characteristics of some nuclei commonly exploited in NMR

Nucleus (spin quantum number, I)	Isotopic abundance	Magnetogyric ratio, γ ($\text{T}^{-1}\text{s}^{-1}$)	Relative sensitivity	Nuclear g-factor (g_N)	Absorption frequency (MHz) at 4.7 T
^1H ($\frac{1}{2}$)	99.98	2.6752×10^8	1.00	5.585	200
^{13}C ($\frac{1}{2}$)	1.11	6.7283×10^7	0.02	1.405	50.3
^{19}F ($\frac{1}{2}$)	100	2.5181×10^8	0.83	5.257	188.2
^{31}P ($\frac{1}{2}$)	100	1.0841×10^8	0.07	2.263	81.0

Besides ^{13}C , two other most widely investigated nuclei are ^{19}F and ^{31}P , both of which not only have $I = \frac{1}{2}$ but both of these nuclei have natural abundance of 100%. In these cases NMR spectra can be analysed in terms of their characteristic chemical shift (δ) and spin-spin coupling constants (J) due to their interaction with other nuclei. In addition to these, other nuclei such as ^2D , ^{11}B , ^{23}Na , ^{15}N , ^{29}Si , ^{109}Ag , ^{199}Hg , ^{113}Cd and ^{207}Pb have also been investigated. NMR of these nuclei is particularly important in the field of Organic Chemistry, Biochemistry, Biology, Organometallic Chemistry, alloys and intermetallic compounds.

12.3 FOURIER TRANSFORM NMR

The intensity of NMR signal is proportional to the concentration of the absorbing species. Many a times we come across situations where the amount of sample available for analysis is too small; especially in case of biological samples. In such cases the signal to noise ratio is very small and it becomes difficult to distinguish the signal from the background noise. A way out is to 'scan' the sample a number of times and add the different scans with the help of a computer. In this process the signal being at the same frequency every time adds up whereas the noise being random in nature gets cancelled out. This improves the signal to the noise ratio, however taking a number of scans is quite time consuming. For example, if one scan can be completed in about one minute, we would need more than an hour to record a meaningful spectrum. A way out is provided by an important process called **Fourier transformation**. Let us learn about this.

Fourier transform (FT) spectroscopy is a general concept used to study very weak signals after isolating it from environmental noise. It was first developed by astronomers in the early 1950s to study the infrared spectra of distant stars. In case of FT-NMR measurements, nuclei in a magnetic field are subjected to very brief pulses of intense radiofrequency radiation. This brief pulse of duration of few microseconds contains all the frequencies in the range where the nuclei can absorb. This causes all the nuclei in the sample to absorb the radiofrequency and get excited simultaneously. After a delay of a few seconds another pulse is applied. The delay is so chosen that during the interval between the two pulses all the excited nuclei relax back to the ground state and the population difference is achieved back. This relaxation is characterised by **free induction decay** (FID). This FID can be detected with a radio receiver coil which is perpendicular to the static magnetic field. In fact, a single coil is frequently used both to provide pulse to the sample and also to detect the FID signal. The FID signal is digitised and stored in a computer for data processing. The process of giving a pulse and collecting the FID takes a few seconds, therefore a large number of such FIDs which are in a way equivalent to a scan can be collected in a reasonable amount of time.

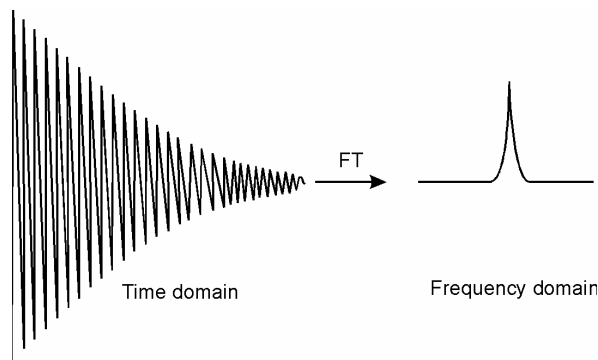


Fig. 12.8: In Fourier transformation the time domain signal (FID) is transformed into a frequency domain signal

The time domain decay signals (FIDs) from numerous successive pulses are added to improve the signal to noise ratio. The resulting summed data are then converted to a frequency domain signal by Fourier transformation and finally, digital filtering may be applied to the data to further increase the signal to noise ratio. The resulting frequency domain spectrum is similar to the spectrum produced by a scanning continuous wave experiment. FT instruments have the advantage of high resolving power and wavelength reproducibility which make possible the analysis of complex spectra.

SAQ 4

In what way FT – NMR is better than CW – NMR?

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12.4 CHEMICAL SHIFT

The description of the behaviour of nuclei in presence of magnetic field discussed so far holds good only for bare nuclei. The resonance condition for such nuclei is, $\nu = g_N \mu_N B_0 / h$. It implies that all the nuclei of a given type (for example, of hydrogen atoms) in a sample should absorb the energy corresponding to the above ν value. If this was the case, the NMR spectroscopy would have been of no use to the chemists. In real systems, however, we do not deal with bare nuclei. The nuclei are surrounded by electrons and their presence can modify the field experienced by the nuclei by either **shielding** or **deshielding** them. This makes the nuclei to come to resonance at different frequencies. Let us understand this.

The electrons circulating around the nucleus (proton in case of H atom) in a spherical fashion produce an **induced magnetic field** (B_{ind}) which opposes the applied field (B_0). Therefore, the magnetic field experienced by the proton in a molecule placed in a magnetic field of strength B_0 is always less because of shielding or screening of the nucleus by the surrounding electrons. The effective magnetic field experienced by the nucleus (B_{eff}) is given by the following equation:

$$B_{\text{eff}} = B_0 - B_{\text{ind}} \quad \dots (12.9)$$

The induced field on the other hand is proportional to the applied field and is given by the following expression:

$$B_{\text{ind}} = \sigma B_0 \quad \dots (12.10)$$

Miscellaneous Methods

A positive value of σ implies that the nuclei are shielded by the electronic environment, while negative σ corresponds to the deshielding of the nucleus.

The proportionality constant, σ , is called the **shielding constant** and is a measure of the extent of shielding of the nucleus by the electrons. Substituting the value of B_{ind} from Eq. 12.10 into Eq. 12.9, we get the following.

$$\begin{aligned} B_{\text{eff}} &= B_0 - \sigma B_0 \\ &= B_0 (1 - \sigma) \end{aligned} \quad \dots (12.11)$$

Thus, in presence of the extra nuclear electronic environment, the resonance condition gets modified as given below.

$$\nu = g_N \mu_N B_0 (1 - \sigma) / h \quad \dots (12.12)$$

In molecular systems containing nuclei other than that of hydrogen and π or conjugated electrons, the field generated by the electrons may augment the applied field. That is the effective field at the nucleus may be more than the applied field. The value of σ will be negative in such a case and we say that the nucleus is deshielded.

When shielding occurs, the B_{eff} is less than B_0 , hence B_0 must be increased to bring the nucleus to resonance. On the other hand when deshielding occurs, B_{eff} is more than B_0 , requiring the field to be reduced to achieve resonance. Thus, the nucleus comes in resonance at lower field. Therefore, due to the shielding (or deshielding) identical nuclei (e.g., H) which have different chemical environment (in other words, different electron density) resonate at different values of the frequencies or applied field. These values being characteristic of the chemical environment can be used to identify various types of environment in which the proton is present. Since the shift in the position of resonance is due to difference in chemical environment, it is called **chemical shift**. Let us learn about the shielding/deshielding mechanism.

12.4.1 Shielding Mechanism

The magnitude of the chemical shift is proportional to the strength of the magnetic field generated by the circulation of surrounding electrons about the proton. In order to understand the chemical shift value originating from structural arrangements, it is essential to understand the nature of this electron circulation and the corresponding induced magnetic field. The protons experience shielding due to a combination of different types of electronic circulations. This is explained as follows.

It may be noted that electronic distribution for a free hydrogen atom and hydride ion are spherically symmetrical. The applied magnetic field induces electronic circulation about the nucleus in a plane perpendicular to the applied field. This generates a small magnetic field which, in the neighbourhood of the nucleus, is opposed to the direction of the applied field as schematically shown in Fig. 12.9. This is called **diamagnetic effect** and causes the resonance to be observed at high field.

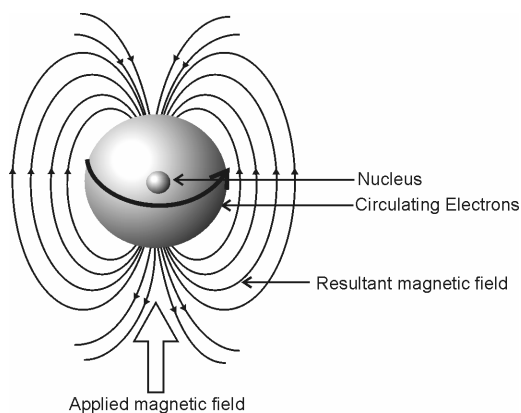


Fig. 12.9: Diamagnetic shielding effect: A mechanism for chemical shift

In organic molecules where hydrogen is bound to carbon and other atoms, the electron distribution is not spherical as in the above case. The electrons occupy *p* and *d* orbitals in addition to the spherical *s* orbitals. In such a case two types of effects are generated. These are diamagnetic effects in which the induced magnetic field direction is opposed to the applied magnetic field and shields the nucleus. The other kind of effect is **paramagnetic effect** and arises due to the induced magnetic field whose direction is parallel to the applied magnetic field. These effects cause deshielding of the nucleus.

12.4.2 Standard for Chemical Shift

As we cannot measure the NMR of bare nuclei, we do not have a reference to measure the chemical shifts. Therefore we need to use some reference standard with respect to which the extent of shielding or deshielding of the external field in various chemical environments can be measured. This is very similar to the choice of the standard hydrogen electrode as the reference for defining electrode potentials of half cells in Electrochemistry. In case of organic compounds generally tetramethylsilane, $(\text{CH}_3)_4\text{Si}$ (TMS) is used as a standard with respect to which chemical shift data is reported. TMS has the following characteristics that make it a molecule of choice to act as a reference.

- It gives a strong and sharp signal at a very high field. Therefore it does not interfere with the signals of all other types of protons in different organic molecules as they absorb at lower field relative to TMS.
- All the 12 protons are chemically and magnetically equivalent therefore, these give a reasonably intense signal even at very low concentrations.
- It has low boiling point (27 °C) and is soluble in most organic solvents. Hence it can be easily removed or separated from other organic compounds or solvents after the spectrum is recorded.
- It is highly inert and does not interact with most organic compounds. Hence it does not interfere in NMR measurements.

12.4.3 Unit of Chemical Shift

You have learnt that in a spectrum the X-axis of the spectrum refers to the energy of the EM radiation in terms of wavelength, frequency or wave number. In case of NMR the X- axis should, in principle, be the frequency of the radiation being absorbed or the field at which the resonance is achieved for a predetermined (fixed) radiofrequency. However, using the frequency or the field is quite inconvenient. More so since the frequency of absorption depends on the applied field, the position of the signals for a given analyte would depend on the applied field. That is the position of the signal obtained for a given analyte on different instruments would be different. Therefore, we need a parameter that is independent of the applied field. Such a parameter is called δ or ppm which is a dimensionless quantity. Let us try to understand the meaning and the genesis of δ .

Suppose we measure a test sample and the reference, TMS using the same magnetic field B_0 , the resonance conditions for the two would be given as below.

$$h\nu_{\text{Test}} = g_N \mu_N B_0 (1 - \sigma_{\text{Test}})$$

$$h\nu_{\text{TMS}} = g_N \mu_N B_0 (1 - \sigma_{\text{TMS}})$$

Thus, the shift in resonance frequency would be

$$h\nu_{\text{Test}} - h\nu_{\text{TMS}} = g_N \mu_N B_0 (1 - \sigma_{\text{Test}}) - g_N \mu_N B_0 (1 - \sigma_{\text{TMS}})$$

$$h(\nu_{\text{Test}} - \nu_{\text{TMS}}) = g_N \mu_N B_0 [1 - \sigma_{\text{Test}} - (1 - \sigma_{\text{TMS}})]$$

$$(v_{\text{Test}} - v_{\text{TMS}}) = \frac{g_N \mu_N B_0 (-\sigma_{\text{Test}})}{h} \quad \because \sigma_{\text{TMS}} = 0$$

Suppose we measure the same two samples (Test and TMS) at double the field then, the shift in resonance frequency would be

$$(v_{\text{Test}} - v_{\text{TMS}}) = \frac{2g_N \mu_N B_0 (-\sigma_{\text{Test}})}{h}$$

That is the frequency difference between test and reference signal gets doubled. Since the shift in resonance position is due to the chemical environment this must be independent of the applied field. This can be achieved by dividing the shift in field (for constant frequency measurement) or shift in frequency (for constant field measurement) by the respective spectrometer field or frequency.

$$= \frac{(v_{\text{Test}} - v_{\text{TMS}})}{v_{\text{op}}}$$

where, v_{op} is the **operating frequency** of the spectrometer.

This makes the shift dimensionless, but a problem still remains. The value of the ratio is very small because the shift in field or frequency is nearly 10^6 times smaller than the measuring field or frequency. Therefore, we multiply this dimensionless quantity by 10^6 and express it in terms of parts per million (ppm). Thus, the chemical shift, δ , is mathematically defined as follows:

$$\delta = \frac{(v_{\text{Test}} - v_{\text{TMS}})}{v_{\text{op}}} \times 10^6$$

Usually, the δ scale ranges from 0 to 12 ppm. A higher value of δ implies the signal to be more downfield from the reference whereas a high field signal will correspond to a lower δ value. In order to circumvent this problem another scale called **τ scale** has been proposed. It is defined as

$$\tau = 10 - \delta$$

Here the TMS is arbitrarily assigned a value of 10 τ . The unit is still in ppm with only difference in scale. The two scales are related as shown in Fig. 12.10.

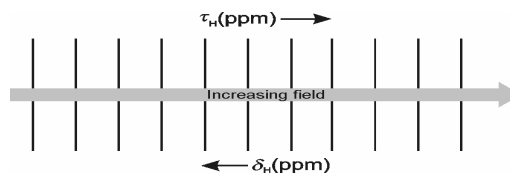


Fig. 12.10: Schematic representation of the relationship between the δ and τ scales of chemical shifts

The typical chemical shift positions of some commonly encountered function groups are given in Fig. 12.11.

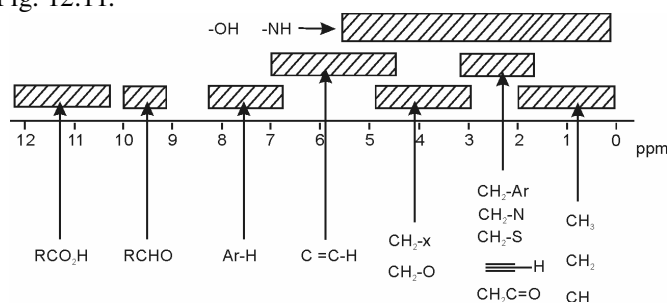


Fig. 12.11: Range of chemical shift values for different types of protons

Commonly, ppm is recognised as a concentration unit which is not the case in NMR and hence should not be confused.

12.4.4 Factors Affecting Chemical Shift

You have learnt that the chemical shift arises due to the two kinds of effects namely the diamagnetic and paramagnetic effects. These in turn arise due to the circulation of the electrons surrounding the nucleus. Therefore, any factor that may alter the electron density in the proximity of nucleus (proton) would affect chemical shift. Let us learn about some of these.

Electronegativity: You know that an atom of high electronegativity in a molecule draws electron density towards itself. This causes a decrease in the electron density leading to deshielding of the nucleus. Thus, with increasing electronegativity δ values will become high or go downfield. Let us see some examples as given in Table 12.2.

Table 12.2: The effect of electronegativity and the number of halogen atoms on the chemical shift position of protons in simple methylhalides

Halogen atom	Electronegativity	Molecule	δ value	Molecule	δ value
F	4.0	CH ₃ F	4.26	CH ₄	0.23
Cl	3.0	CH ₃ Cl	3.10	CH ₃ Cl	3.10
Br	2.8	CH ₃ Br	2.65	CH ₂ Cl ₂	5.33
I	2.5	CH ₃ I	2.10	CHCl ₃	7.24

The data shows that the **effect of increasing electron withdrawal on the chemical shift of the remaining protons is cumulative but not additive**. It may be concluded that with increasing electronegativity, shielding of protons decreases.

Anisotropy of chemical bonds: Chemical shift is dependent on the orientation of the NMR active nucleus with respect to the neighbouring bonds especially the π bonds. In such cases the circular motion of π electrons in the presence of applied magnetic field generates induced magnetic field which is anisotropic in nature. **Anisotropic** means that for some part of the molecule the field opposes the applied field and for other parts it augments the applied field. Let us understand this with the help of the example of carbonyl group. The induced magnetic field for this group is shown in Fig. 12.12.

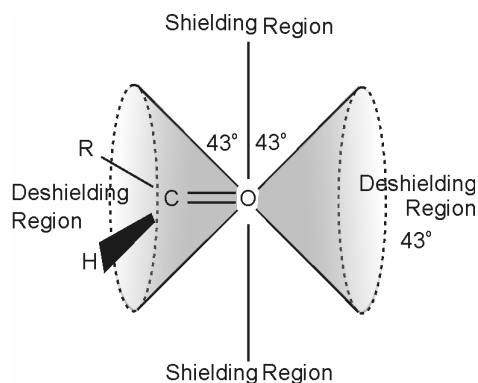


Fig. 12.12: Anisotropic shielding and deshielding around a carbonyl group

There are two cone shaped volumes that lie parallel to the C=O bond axis. These are the deshielding regions and any proton falling in these regions would come to resonance at low fields or high δ value. The high chemical shift (~ 9.2 ppm) of the aldehydic protons is a typical example. The protons that are outside the region of these cones would be shielded from the applied field and accordingly come to resonance at high field.

In case of alkynes or acetylenic protons, however, the protons appear upfield in the range 1.5 to 3.5. Electron circulation around the triple bond occurs in such a way that protons experience a diamagnetic shielding effect as shown in Fig. 12.13. In this case, it is essential to consider the parallel as well as perpendicular orientations of acetylene molecule. If the axis of acetylene molecule is aligned parallel to the applied magnetic field B_0 as shown in Fig. 12.13(b), the π -electrons in the triple bond readily induce diamagnetic circulation and the magnetic field so generated opposes the applied field at the acetylenic protons. This is in addition to the local diamagnetic shielding effects experienced by protons. However, if the axis of acetylene molecule is perpendicular to the applied field as shown in Fig. 12.13 (a) then π -electron circulation is severely restricted and as can be seen from the figure, the induced field augments the applied field at the acetylenic protons. It is because of this pronounced anisotropy of the triple bond, the additional shielding, experienced by the protons varies considerably with the orientation. The net effect of the two effects is that the diamagnetic anisotropy of the triple bond predominates and serves to increase the shielding of the acetylenic protons. This explains the high field shift of the acetylenic protons.

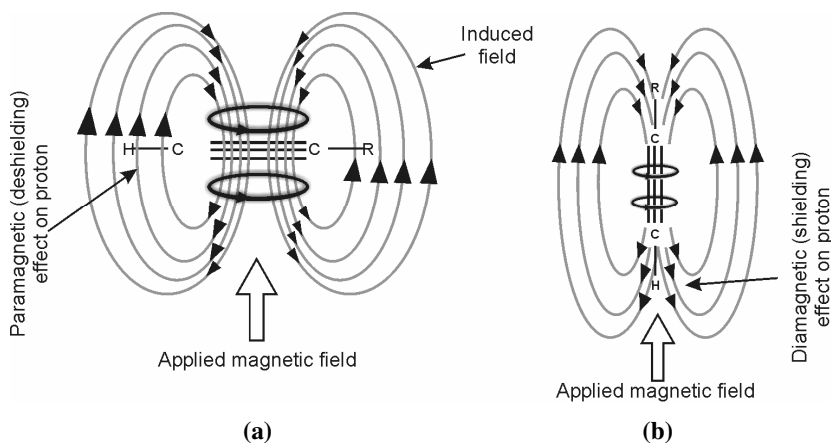


Fig. 12.13: Shielding of acetylenic protons by triple bond in (a) perpendicular to the applied field and (b) parallel orientation

In aromatic rings such as benzene, a different type of diamagnetic shielding is observed due to the unsymmetrical distribution of π -electrons. These are readily induced into circulation in the plane of the ring by the applied magnetic field. Fig 12.14 shows the secondary magnetic field generated by the induced circulation of π -electrons in benzene molecule aligned perpendicular to the applied field. The effects of the secondary magnetic field on a rigidly attached proton in the molecule do not average to zero for all possible orientations of the ring with respect to the applied field. You may note here that secondary magnetic field causes pronounced shielding at the centre of the ring but deshielding outside, in the plane of the ring containing protons.

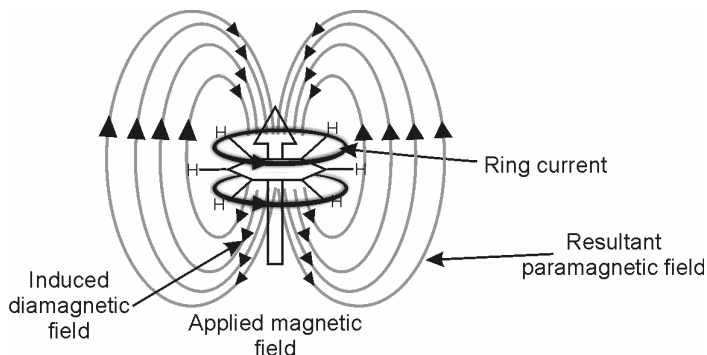


Fig. 12.14: Deshielding of aromatic protons due to induced magnetic field

Due to the pronounced deshielding the δ value for benzene protons is observed at 7.22 ppm, however, it gets shifted to lower or higher side in the range 6.5 to 8.5 ppm, depending on the nature of substituent, in the benzene derivatives. In fact a chemical shift value in this range is an indicator of the presence of benzene or aromatic compounds.

Hydrogen bonds: In case of hydrogen bonding, the H-atom is shared by two electronegative atoms. It is highly deshielded and the resonance occurs at low field and correspondingly high δ value is observed. All alcohols, amines and thiols show varying degree of H-bonding.

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State whether the statements given below are true (T) or false (F).

- i) The magnitude of the chemical shift is proportional to the strength of the induced magnetic field generated by the circulation of surrounding electrons about the protons.
- ii) Tetramethylsilane, used as a standard for measuring chemical shifts in organic compounds interferes with the signals of other types of protons in the organic molecules.
- iii) The acetylene molecule shows a pronounced diamagnetic as well as paramagnetic anisotropic effect however, the diamagnetic anisotropic effect predominates causing shielding of the acetylenic protons.

12.5 SPIN-SPIN COUPLING

Under low resolution *i.e.*, at low fields we generally get as many signals in the NMR spectrum as are the different types of protons in the molecule. The ratio of intensities of the NMR signals as measured by the areas under the peaks is a measure of the number of protons in each group. However, at high resolution, spectral bands of a molecule containing nonequivalent protons are split into two or more components depending on the nature of neighbouring groups. The splitting of the signals is due to the interaction of nuclear spins of adjacent nuclei and the phenomenon causing such splitting of the signals is called **spin-spin coupling**. The interaction or the coupling of the spins is through the bonding electrons. This effect is normally not observed if the coupled protons are more than three σ bonds away. The splitting pattern of different signals can provide very important structural information. Let us learn about it by taking the example of ethyl chloride.

Under the conditions of low resolution, ethyl chloride shows two signals in its NMR spectrum one each corresponding to $-\text{CH}_3$, and $-\text{CH}_2$ protons. The peak areas are in the ratio of 3:2. When the same spectrum is recorded with a high resolution spectrometer, CH_3 absorption band splits into three lines (called triplet) and CH_2 band splits into four lines (a quartet). This splitting can be explained in terms of the changes in magnetic field experienced by the protons of one group due to the possible spin arrangements of the protons in the adjacent groups.

Let us focus our attention on the methyl protons of ethyl chloride. In the absence of the neighbouring methylene group protons the methyl protons would come to resonance at a certain frequency depending on the effective field experienced by them. It will give a single signal in the low resolution spectrum. In the presence of methylene group with two equivalent protons the field experienced by the methyl protons gets altered because the nuclear spins of methylene protons also act as bar magnets. The magnitude of the effect would depend on the relative orientations of the nuclear spins of these protons. The nuclear spins of the two protons can have four

possible combinations of spin orientations in the applied magnetic field as shown in Fig. 12.15.

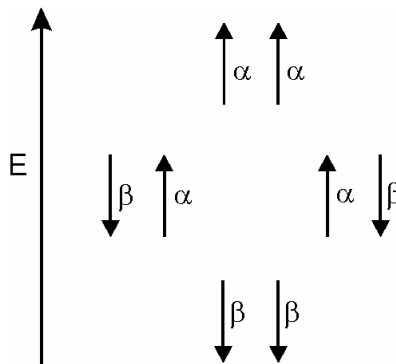


Fig. 12.15: Possible spin orientations of two protons of the methylene group

In the first combination both the spins are in the α state. As these are aligned with the direction of the applied field, these would augment the magnetic field experienced by the methyl protons. As a consequence, the methyl protons would come to resonance at a lower field than that in the absence of methylene protons. Similarly, the fourth combination in which both the spins are in the β state would make the methyl group protons to come to resonance at a higher field. The other two combinations with one α and one β spins would have no net effect on the methyl protons absorption. Therefore, the resonance absorption signal of the methyl protons would be split up into three peaks (triplet) having relative areas in the ratio of 1:2:1. We are sure that you can justify the ratios of 1:2:1.

In the same way if we focus on the methylene protons and look for the possible spin combinations of the methyl group protons we find that there are eight such combinations (Fig. 12.16). As you can notice, these combinations fall in four groups. Accordingly, the methylene protons signal splits into four lines (quartet) having an intensity ratio of 1:3:3:1

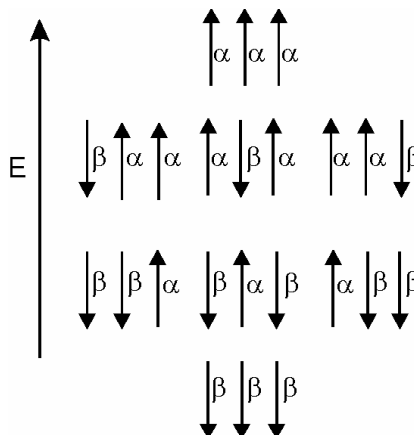


Fig. 12.16: Possible spin orientations due to three protons of methyl group

Similarly, we can take other examples of molecules in which a given type of proton has one, two, three or more equivalent neighboring protons. It has been found that the number of lines in a signal (multiplicity) is governed by $(n+1)$ rule. A given signal would split into $n+1$ lines where n is the number of equivalent protons on the adjacent C atom(s).

In simple cases of interacting nuclei, the relative intensities of the lines in a multiplet is given by the coefficients of the terms in the expansion of $(1+x)^n$. This can also be represented as **Pascal's triangle** shown in Fig. 12.17.

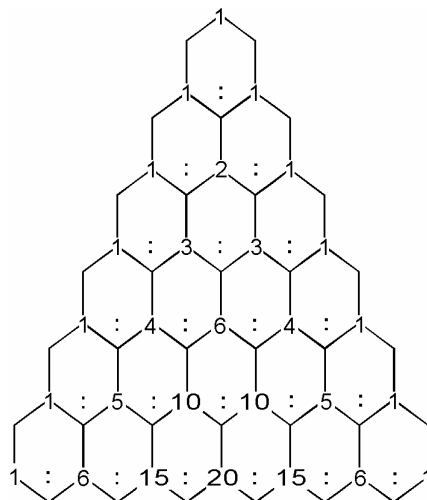


Fig. 12.17: Pascal's triangle showing relative areas/intensities of split lines

The separation of these peaks in frequency units is called **coupling constant**. It is denoted by J and is a measure of the strength of the coupling interaction. The coupling constant is independent of the strength of the applied magnetic field.

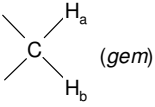
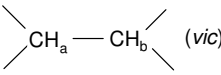
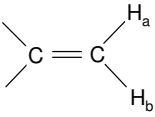
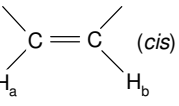
These simple rules for determining the multiplicity of spin-spin interactions of adjacent groups hold only for cases where the separation of resonance lines of the interacting groups (Δ) is much larger than the coupling constant (J) of the groups ($\Delta \gg J$). If $\Delta \approx J$ then the simple rules of multiplicity no longer hold good.

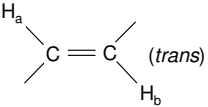
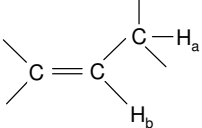
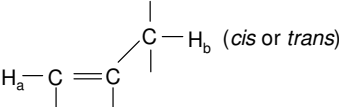
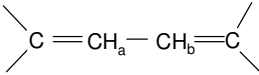
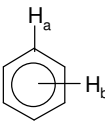
12.5.1 Magnitude of Coupling Constants

The magnitude of the spin-spin coupling constant is found to depend on the relative orientation and the distance of the interacting nuclei. On the basis of the data on the coupling constants of a large number of molecules different types of couplings have been identified.

Different types of spin-spin couplings and the values of their magnitudes are compiled in Table 12.3.

Table 12.3: Magnitude of different spin-spin coupling constants for protons

Function	J_{ab}/Hz
 (gem)	10-18 depending on the electronegativities of the attached groups
 (vic)	Depends on dihedral angle
 (trans)	1-4
 (cis)	5-14

	11-19
	4-10
	0-2 (for aromatic systems, 0-1)
	10-13
	{ <i>ortho</i> 7-10 <i>meta</i> 2-3 <i>para</i> 0-1

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Tick mark (✓) for the options given below which hold right for the coupling constant.

The coupling constant depicting the separation of peaks obtained in NMR spectra is:

- i) measurement of the strength of the coupling interaction.
- ii) dependent on the strength of the applied magnetic field.
- iii) equal to 11-19 Hz for *trans* coupling and 5-14 for *cis* coupling.
- iv) independent of the relative orientation and the distance of the interacting nuclei.

12.6 INSTRUMENTATION FOR NMR SPECTROSCOPY

Unlike other spectrometers, an additional device – a magnet capable of producing strong homogenous magnetic field is needed in NMR spectrometer. The essential components of NMR instrument include the following.

- a highly stable magnet
- sample probe
- source of radio frequency radiation
- phase sensitive detector
- data processing unit

Schematic block diagram of a typical NMR spectrometer illustrating various components is shown in Fig. 12.18.

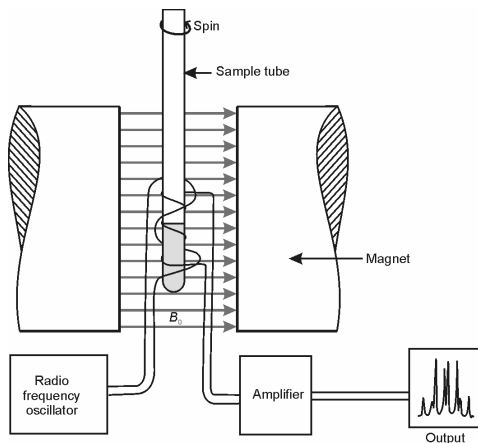


Fig.12.18: Schematic diagram of NMR spectrometer showing various components

Let us study about the different components of NMR spectrometer.

12.6.1 Magnet

It is the heart of all types of NMR spectrometers. The sensitivity and resolution of a spectrometer critically depends on the strength and quality of the magnet. Since sensitivity and resolution both increase with field strength, it is advantageous to operate the instrument at highest possible field strength. In addition, the field must be homogeneous, uniform and reproducible. Three types of magnets have been used in NMR spectrometers; these are permanent magnet, electromagnet and super conducting solenoid.

Permanent magnets with strengths that need an oscillator frequency of <100 MHz for bare protons have been used in commercial continuous wave spectrometers. These are very temperature sensitive and require extensive cooling and shielding. However, these are not ideal for extended periods of data accumulation because of field drift problems. The electromagnets are now rarely used in the NMR instruments. The modern high resolution spectrometers use superconducting magnets of T or above. These are simple, small sized and produce high field strength besides having low operating cost. It is very essential that stability of a magnet is maintained at all costs.

Tesla: It is the SI unit of magnetic flux density named after Serbian-American physicist Nikola Tesla.

$$1\text{T} = 10^4 \text{ gauss}$$

12.6.2 The Sample Probe

It is a key component in NMR spectrometer which not only holds the sample in a fixed position in the magnetic field but also contains an air turbine to rotate (spin) the sample. The sample is spun so as to ensure that it experiences a uniform field. In addition to these, it houses a coil for generating radiofrequency for excitation and also for detection of the NMR signal. Earlier NMR spectrometers used to have separate transmitter and receiver coils perpendicular to each other for excitation of the pulse and detection of the signal. Modern spectrometers contain a single coil probe which is simple and more efficient. Since most of the continuous wave spectrometers use field sweep mode, a fixed frequency is generated from the radiofrequency generator and frequency synthesizer. The spectrometer is known by its operating frequency e.g., a 60 MHz, a 100 MHz or a 500 MHz instrument and so on. However, in Fourier transform spectra, the sample is irradiated with a pulse consisting of a range of frequencies sufficiently great to excite nucleus having different resonance frequencies. The pulse of radiation provides a relatively broad band of frequencies centered on the oscillator frequency.

12.6.3 Detector System

High frequency radio signal is first converted to an audio frequency signal which can be thought of as being made up of two components; a carrier signal which has the

frequency of the oscillator that is used to produce it and a superimposed NMR signal from the analyte. The analyte signal differs in frequency from the standard by a few ppm. In case of proton spectrum, chemical shifts are typically in the range of 1-10 ppm. Thus proton magnetic resonance data generated by 200 MHz spectrometer would lie in the frequency range 200,000,000 Hz to 200,002,000 Hz. It is impractical to digitise such a small difference. In practice, a difference signal is obtained that lies in the audio frequency in kilohertz range. The modern spectrometers contain a quadrupole phase sensitive detector which is capable of sensing the sign of frequency difference. This allows the determination of positive and negative difference between the frequency of standard and sample.

12.6.4 Sample Handling

In general, high resolution NMR spectroscopy work requires clear transparent sample solution of 2 to 15% concentration. However, pure liquids may also be used provided these are not viscous or have low viscosity. The sample is taken in a glass tube of approximately 5 mm outer diameter and 15-20 cm length capable of containing 0.5 cm³ liquid sample. For small size samples, micro tubes are also available.

An important aspect of sample handling is that the solvents should be aprotic in nature. This is so because if the solvent contains hydrogen atom then the absorption by the solvent would interfere with that of the analyte. More so, as the amount of the solvent is much more than the analyte, its signal would be much larger than that of the analyte. This may sometimes mask the analyte signal. Carbon tetrachloride which does not contain any hydrogen atom is considered as the most ideal. However, low solubility of many compounds in carbon tetrachloride puts a limitation on its use. In order to avoid this problem, a variety of deuterated organic solvents are used. Most commonly used solvents are; deuterated chloroform (CDCl₃), deuterated benzene (C₆H₆), deuterated acetone (CD₃COCD₃) and deuterated dimethylsulfoxide (CD₃SOCD₃) etc.

12.6.5 Representation of NMR

The NMR spectrum is recorded on a chart paper with X-axis representing chemical shift (δ) in ppm and Y-axis as intensity. The δ values increase from right to left. The value of zero on the X axis corresponds to the internal standard, tetramethylsilane (TMS) signal. As you have learnt earlier, the low δ values correspond to high field and vice versa. The NMR spectra are normally recorded in two modes; absorption and **integral modes**. The output consists of two traces – one is the spectrum i.e., the **absorption signals** at different δ values and the other called **integration trace**.

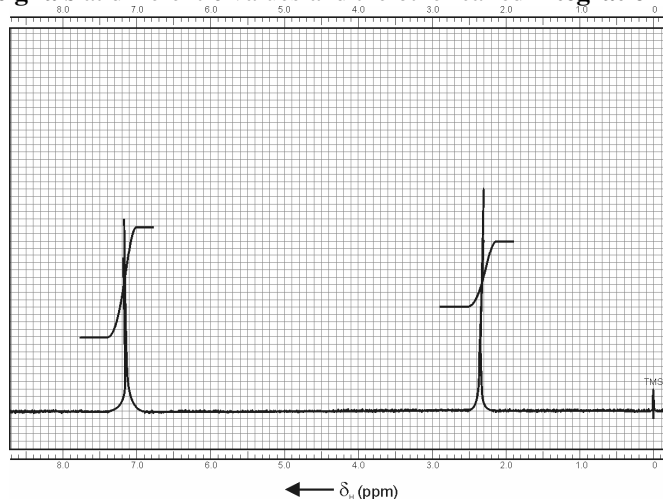


Fig. 12.19: A sample NMR spectrum showing the absorption and the integration traces

The integral trace gives the area under different peaks and appears as step function superimposed on the NMR spectrum as typically illustrated in Fig. 12.19. The integral tracing is usually recorded from left to right. The height that the tracing rises for each group of protons is proportional to the area enclosed by the peak, and therefore to the number of protons responsible for that absorption.

In general, however, NMR spectra can be classified into two broad groups; **low resolution or broad / wide line** and **high resolution or sharp line spectrum**. In a wide line spectrum, band width of the lines is large enough and the fine structure due to chemical environment causing spin-spin splitting is obscured. Such spectra are obtained from low field NMR spectrometers usually of < 100 MHz and are useful for the study of physical environment of the absorbing species.

SAQ 7

Why should we use aprotic solvents to record NMR spectrum of a compound?

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12.7 APPLICATIONS OF NMR SPECTROSCOPY

By far NMR spectroscopy is most widely and routinely used for the identification and structure elucidation of organic, organometallic and biological molecules. However, very few attempts have been made for quantitative determination of absorbing species by NMR. This is because of high cost factor involved whereas the same information could be obtained from other economical techniques. In addition, overlap of many resonance peaks in complex systems makes the job further difficult. Despite these limitations NMR can be conveniently used for quantitative analysis. Let us take up some examples of quantitative analysis before we take up identification and structural analysis.

12.7.1 Quantitative Applications

NMR spectrum has a unique feature of direct proportionality between peak area and the number of nuclei responsible for that peak. Unlike other spectroscopic techniques where a calibration plot is prepared by using a pure compound of varying concentrations, no pure sample is required for calibration in NMR. Thus, if an identifiable peak for one of the constituents of a sample does not overlap with the other constituent peaks, the area of this peak can be used to establish the concentration of the species directly provided that signal area per proton is known. The signal area per proton, on the other hand, can be calculated by using a known concentration of an internal standard. It may be emphasised that the peak of the internal standard should not overlap with any of the sample peaks. In this regard organic silicon compounds are unique for calibration purpose because of high up field location of their proton peaks.

Some examples of quantitative determinations by NMR are:

Analysis of Multicomponent Mixtures: A mixture of aspirin, phenacetin and caffeine in commercial analgesic preparation can be analysed by NMR with a relative error of 1 to 3%. Similarly a mixture of benzene, heptane, ethylene glycol and water can be precisely and rapidly determined by NMR.

Elemental Analysis: Though NMR is not routinely used for the determination of total elemental concentration yet, in principle it could be used for given NMR active nucleus in a sample. The integrated NMR intensities of the proton peaks for a large number of organic compounds can be used for the accurate quantitative determination of total hydrogen atoms in an organic mixture.

Organic Functional Group Analysis: A useful application of NMR is the determination of functional groups such hydroxyl groups ($-OH$) in alcohols (ROH) and phenols (C_6H_5OH), aldehydes, carboxylic acids, olefinic and acetylenic hydrogens, amines and amides with relative error of 1 to 5% range.

12.7.2 Qualitative Applications

Literature is full of references illustrating identification of organic compounds and their structure determination by NMR spectroscopy. We shall discuss the spectral characteristics of some typical examples of simple organic compounds. We have chosen examples of aliphatic and aromatic compounds containing a hydroxyl group and would like to demonstrate some structural and dynamic features that may be studied by NMR. We begin with a simple molecule, ethanol.

Ethanol is a classical example of identification, spin-spin coupling and structure determination including exchange phenomenon of an organic compound. A low resolution NMR spectrum of ethyl alcohol (Fig. 12.20) consists of three broad peaks corresponding to three chemically different types of protons with peak areas in the ratio of 3:2:1. It suggests for the presence of three protons of one type, two of another type and one of third type which correspond to methyl ($-CH_3$), methylene ($-CH_2-$) and an alcoholic ($-OH$) group respectively. Of these, $-OH$ proton is observed at most downfield because it is attached to an electronegative oxygen atom and the one corresponding to methyl group is observed high field and has peak area three times that of hydroxyl proton.

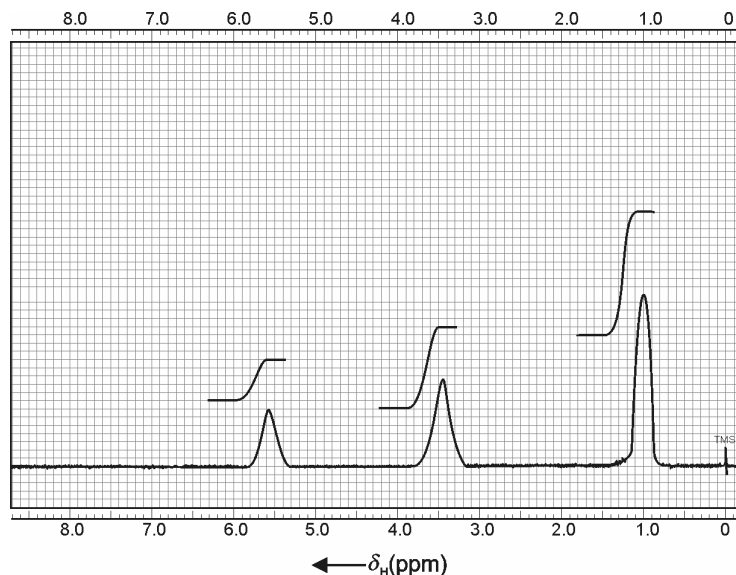


Fig. 12.20: Low resolution NMR spectrum of pure ethanol, CH_3-CH_2-OH

Under high resolution (Fig 12.21) the absorption peaks due to methyl and methylene protons appear as triplet and quartet respectively where total areas are still in the ratio of 3:2. The appearance of triplet and quartet may be understood in terms of $(n+1)$ rule where methyl protons interact with two protons of methylene group giving rise to $2+1 = 3$ lines (triplet) in relative area ratio of 1:2:1. The methylene group protons interact with three protons of methyl group giving rise to $3 + 1 = 4$ lines (quartet) in

relative area ratio of 1:3:3:1 suggesting absence of any spin-spin interaction with the neighbouring hydroxyl group, which appears as a singlet.

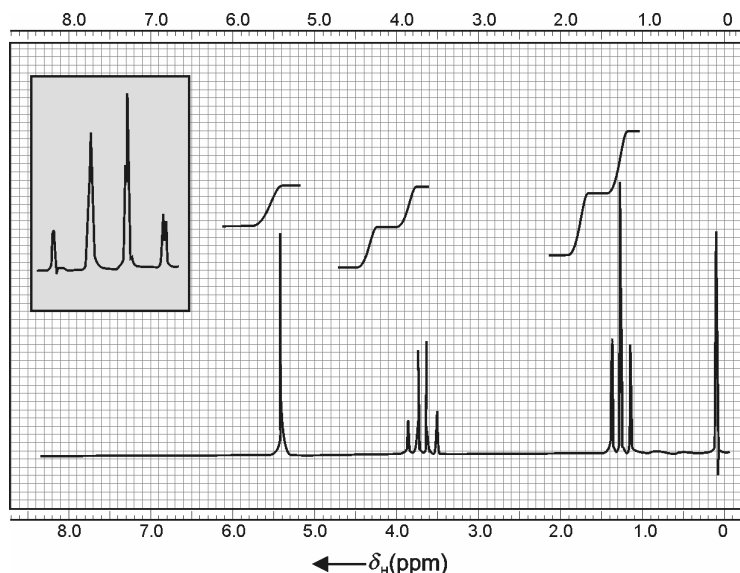


Fig. 12.21: NMR spectrum of pure acidified ethanol at high resolution

However, if the spectrum of highly purified ethanol sample is examined as shown in Fig. 12.22, the hydroxyl proton signal splits into a triplet due to spin-spin interaction with two methylene protons. Further, the multiplicity of methylene protons is also increased; each of four lines gives rise to a doublet due to interaction with the hydroxyl proton. The expanded spectrum in the region of the methylene protons is shown in the inset of Fig. 12.22.

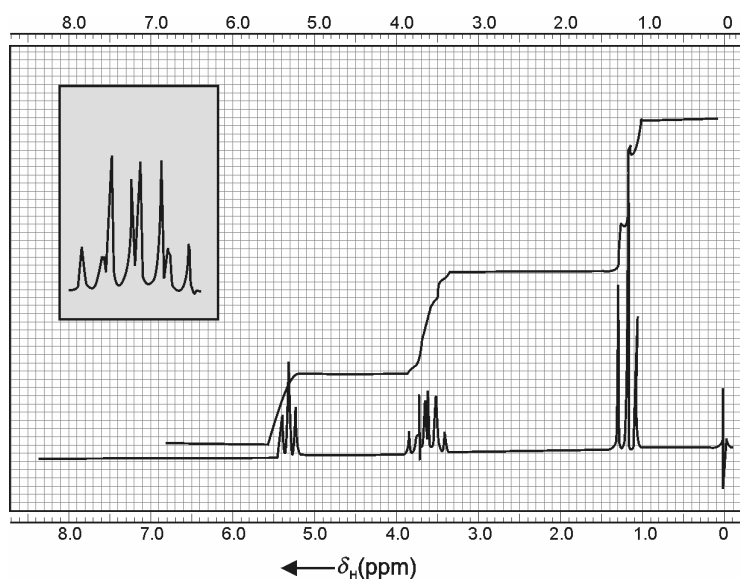


Fig. 12.22: NMR spectrum of highly purified ethanol under high resolution

The difference in the multiplicity of hydroxyl proton in pure and acidified alcohol samples can be best explained in terms of **chemical exchange**. In a given period of time, a single hydroxyl proton may be attached to a number of different ethanol molecules. The rate of proton transfer in pure ethanol is relatively slow and the proton is available on the oxygen atom and can interact with the methylene protons giving a triplet. However, in presence of acidic or basic impurities ordinarily present in the sample the rate of exchange is markedly increased and the proton does not reside on

the oxygen atom for sufficiently long to interact with methylene group and only a sharp singlet is observed as shown in Fig. 12.21.

Methanol is a very simple case as far as NMR spectrum is concerned. It is a good example to demonstrate chemical exchange at high temperatures. At room temperature, NMR spectrum of methanol ($\text{CH}_3\text{-OH}$) as expected, exhibits two lines. A sharp signal at high field is due to methyl group (CH_3) and a low intensity downfield signal is due to alcoholic proton (OH). However, on lowering the temperature of measurement it shows interesting changes that are attributed to the exchange phenomenon. The NMR spectra of CH_3OH at varying temperatures of 31, 6, 1, -4 , -6 , -14 , and -40°C are shown in Fig. 12.23.

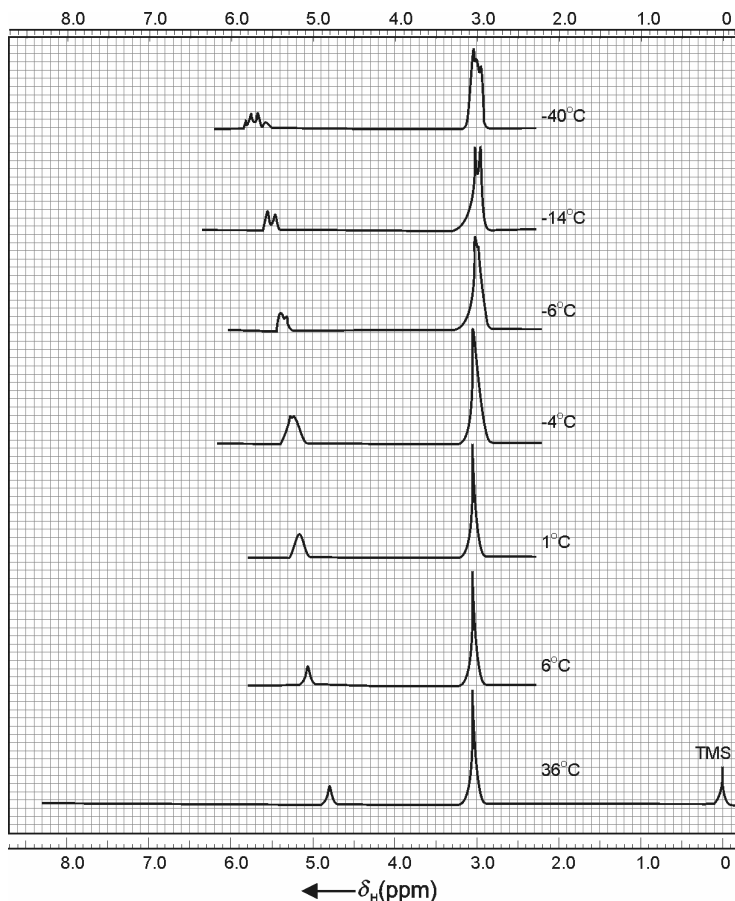


Fig. 12.23: NMR spectra of methanol at varying temperatures

You may note that as the temperature is lowered from 31°C to 6°C and then to 1°C no splitting occurs in any of the two lines though both lines get broadened; the broadening in OH signal being more prominent. On further lowering the temperature down to -4°C broadening becomes more significant suggesting some kind of interaction which, however, becomes clear at -6°C . At -14°C , methyl group signal shows a doublet ($J = 5.2\text{ Hz}$) due to spin-spin interaction with OH proton. At the same time OH signal also shows a doublet but with some structure on both sides. On further lowering of temperature to -40°C , a clear quartet ($J = 5.2\text{ Hz}$) is observed for the hydroxyl proton.

This kind of observation can be explained in terms of chemical exchange. The observed spectra indicate that at room temperature rate of chemical exchange is very fast giving rise to two sharp singlets. However, at -40°C , rate of chemical exchange is very slow giving rise to multiplets arising out of spin-spin coupling.

Benzyl alcohol: In order to understand the effect of benzene ring in alcohols, let us take the example of benzyl alcohol. The spectrum of benzyl alcohol recorded at 60 MHz is shown in Fig. 12.24. In the two cases discussed above you have seen that OH signal is observed at down fields due to O atom being electronegative. However, in the case of $C_6H_5CH_2OH$, the OH signal is observed quite upfield ($\delta = 2.5$ ppm) and the benzyl group downfield ($\delta = 7.2$ ppm) with methylene group ($-CH_2-$) in between the two

($\delta = 4.6$ ppm). You know that the downfield shift of benzyl group protons is due to the anisotropic effect of the benzene ring. In none of these cases, however, any multiplicity is observed.

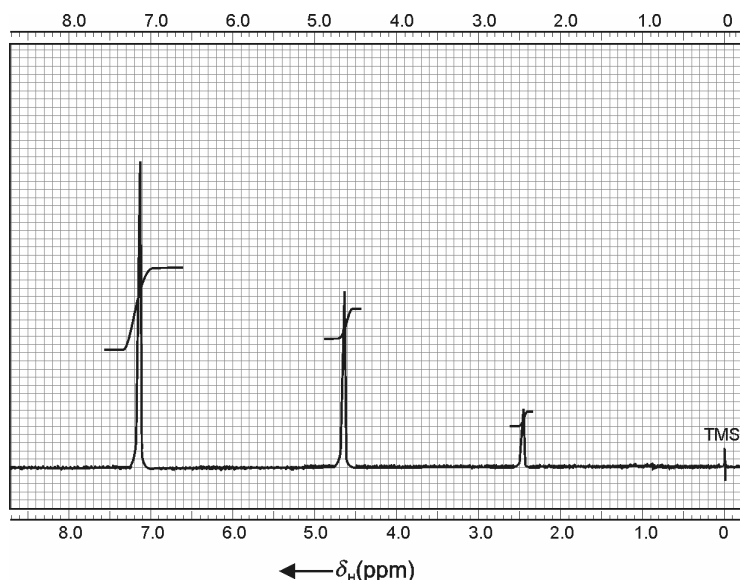


Fig. 12.24: NMR spectrum of benzyl alcohol ($C_6H_5CH_2OH$) in CCl_4 at low resolution

Phenol is another typical case because of the acidic nature of the $-OH$ group unlike other hydroxyl compounds discussed above. The nature of spectrum is strongly concentration dependent. At ordinary concentration, a strong sharp signal is observed in the range 6.0 to 7.7 ppm. However, at low concentration or for dilute solutions, signal is shifted upfield in the range 4 to 5 ppm. Variation of phenolic hydroxyl signal with concentration is shown in Fig. 12.25.

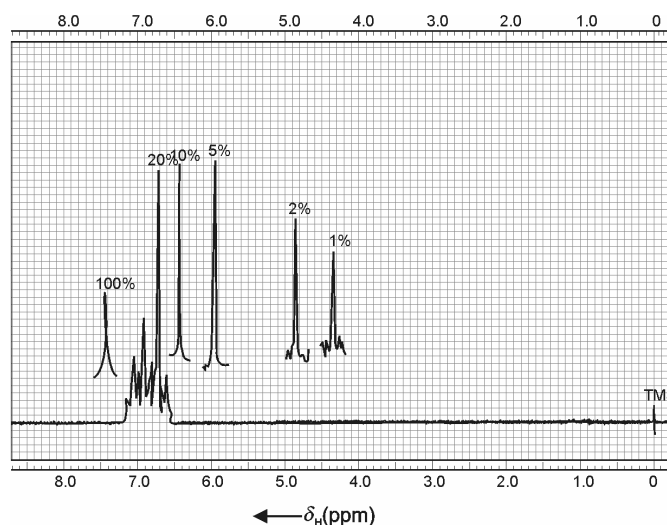


Fig. 12.25: NMR spectrum of phenol, 20% (w/v) in CCl_4 . Also shown is variation of hydroxyl absorptions at various concentrations for pure, 10%, 5%, 2% and 1% solutions

The hydroxyl signal for pure phenol is observed at 7.45 ppm which at 20% (w/v) in carbon tetrachloride is shifted to upfield at 6.75 ppm. The small peaks observed by the side of –OH signal may be attributed to aromatic benzene protons. On further dilutions at 10, 5, 2 and 1% concentrations, –OH signal is continuously shifted to upfield at 6.45, 5.95, 4.88 and 4.37 ppm respectively. This shift can be attributed to loss of intermolecular hydrogen bonding on dilution.

Having learnt about the phenomenon of NMR, the characteristics of NMR spectrum, the instrumentation required to record the spectrum and the applications of NMR let us sum up what all have we learnt in this unit. However, why don't you assess your understanding by solving the following SAQ before that?

SAQ 8

How does the rate of chemical exchange affect the appearance of NMR spectrum?

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12.8 SUMMARY

Nuclear magnetic resonance (NMR) spectroscopy is concerned with the transitions between different nuclear spin states of certain nuclei obtained by placing the sample in an external magnetic field. The information obtained from NMR spectrum can be used to decipher the structure of molecules.

Any nucleus with a spin angular momentum quantum number, $I \neq 0$, corresponds to a spinning positive charge and is associated with a magnetic moment (μ) i.e., these act as tiny bar magnets. When placed in an applied field, these nuclei can take up any of the $(2I + 1)$ orientations with respect to the direction of the applied field. For a nucleus with $I = \frac{1}{2}$ it can take up only two orientations. These are aligned either with or against the field; these correspond to two different energy levels. The available spins distribute themselves between these spin states according to the Boltzmann distribution. In addition to aligning with or against the field these also start precessing around the direction of the applied field. The energy separation of the two levels and the frequency of precession are proportional to the applied field. Incidentally, the frequency required causing a transition between these levels and the precessional frequency is identical.

The transition from the spin state of lower energy to the one of higher energy is brought up with the help of a radiation in the radiofrequency region. The transition called spin flip occurs when the radio frequency and the precessional frequencies are matched. This condition is called resonance. Once excited, the spins relax down to the lower state through radiationless processes called relaxation. There are two types of relaxations namely, spin-lattice relaxation and spin-spin relaxation. In an another method called FT-NMR the nuclei in a magnetic field are subjected to a pulse of intense radiofrequency radiation of a few microseconds duration. This causes all the nuclei in the sample to absorb the radiofrequency and get excited simultaneously. The relaxation is characterised by a free induction decay (FID) which is detected and Fourier transformed to obtain a signal in the frequency domain.

Different protons present in a sample come to resonance at different frequencies depending on their chemical nature. The shift in the frequency of a given proton with respect to an internal standard is called chemical shift and is measured in terms of a dimensionless parameter called δ or ppm. The chemical shift position of a proton depends on a number of factors like the electronegativity of the attached groups; the anisotropy of the bond, hydrogen bonding, etc. The signal for a given kind of proton is split into a number of lines. The number of lines and their intensity ratios depend on the number of equivalent neighbouring protons. This is called spin-spin coupling and the magnitude of the splitting called coupling constant is a measure of the energy of interaction between the protons.

The NMR spectroscopy is very widely and routinely used for the identification and structure elucidation of organic, organometallic and biological molecules. However, very few attempts have been made for quantitative determination of absorbing species also. Some of the quantitative applications of NMR spectroscopy are: analysis of multi-component mixtures, elemental analysis and organic functional group analysis.

12.9 TERMINAL QUESTIONS

- What are the advantages of using a magnet with higher field strength in NMR spectroscopy?
- Calculate the value of nuclear Bohr magneton, μ_N .
- Calculate the Larmor frequency for a proton kept in an external magnetic field of 7.1 T.
- Predict the nature of high resolution proton magnetic resonance spectra of following molecules.
 - Toluene, $C_6H_5CH_3$
 - Ethanoic acid, CH_3COOH
 - Methyl isopropyl ketone, $CH_3CO-i-C_3H_7$
- Explain the following terms.
 - Shielding constant
 - Anisotropy of chemical bond
 - Larmor frequency
 - Coupling constant
- NMR spectrum of an organic compound with molecular formula $C_4H_7BrO_2$ recorded using TMS as a standard gives a triplet, a quartet and a triplet at $\delta = 1.2, 2.1$ and 4.2 ppm respectively. In addition, a sharp peak is observed at $\delta = 11$. Predict the structure of the compound.
- Explain how lines due to chemical shift may be differentiated from those due to spin-spin splitting.

12.10 ANSWERS

Self Assessment Questions

- The following nuclei have $I \neq 0$
 $^2D, ^6Li, ^{11}B, ^{15}N, ^{19}F, ^{23}Na, ^{27}Al, ^{31}Si, ^{31}P, ^{37}Cl, ^{39}K$

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Therefore, these would have magnetic moments and be NMR active. Out of these ^2D , ^6Li would have $I = 1$ and rest would have $I = \frac{1}{2}$

2. According to the Boltzmann distribution, $\frac{n_2 \text{ (upper)}}{n_1 \text{ (lower)}} = e^{(-\Delta E/kT)}$

$$\begin{aligned}\Delta E &= g_N \mu_N B_0 \\ &= 1.405 \times 5.05 \times 10^{-27} \text{ J T}^{-1} \times 2.3 \text{ T} \\ &= 1.92 \times 10^{-26} \text{ J.}\end{aligned}$$

As $\Delta E/kT$ is very small, we may write $e^{(-\Delta E/kT)}$ to be approximately $(1 - \Delta E/kT)$.

This gives

$$\begin{aligned}n_2/n_1 &= 1 - (1.92 \times 10^{-26} / 1.380658 \times 10^{-23} \times 300) \\ &= 1 - 4.635 \times 10^{-3} \\ &= 0.99536\end{aligned}$$

3. i) magnetogyric ratio
ii) continuous wave spectrometry
iii) spin-lattice relaxation, spin-spin relaxation
iv) spin-lattice
4. FT – NMR provides for a quick method of getting good signal to noise ratio in the spectrum even for very small amount of the analyte. Further, FT instruments have the advantage of high resolving power and wavelength reproducibility which make possible the analysis of complex spectra.
5. i) T ii) F iii) T
6. i) and iii)
7. We should use aprotic solvents to record NMR spectrum of a compound because the signals of the hydrogen atoms present in the solvent may interfere with that of the analyte. Further, since the amount of the solvent is larger than that of the analyte its signal may sometimes mask the analyte signal.
8. The rate of chemical exchange for the exchanging proton determines whether or not the proton is available for coupling interactions with the neighbouring protons. The appearance of the spectrum depends on the rate of chemical exchange. The molecules under the condition of slow chemical exchange would undergo spin-spin splitting and give spectra with splitting pattern. On the other hand when the conditions allow fast chemical exchange, the coupling interactions are lost and we get simplified spectrum. Thus, in order to observe the coupling interactions the exchange rate must be very small.

Terminal Questions

1. The chemical shift is defined as, $\delta = \frac{(\nu_{\text{Test}} - \nu_{\text{TMS}})}{\nu_{\text{op}}} \times 10^6$ and is characteristic of the nature of the proton. If we use a magnet of high field strength (which would mean a higher operating frequency), the shift in the signal of the test proton from that of the reference proton would be larger. This helps in the resolution of the signals.

2. The Bohr magneton is defined as $\mu_N = \frac{e\hbar}{2m_p}$ where e is the electron charge, $\hbar$ is the action constant $= h/2\pi$; h being Planck's constant and m_p is the mass of the proton.

Substituting the values we get,

$$\mu_N = \frac{e\hbar}{4\pi m_p}$$

$$\mu_N = \frac{1.602 \times 10^{-19} \text{ C} \times 6.626 \times 10^{-34} \text{ Js}}{4 \times 3.141 \times 1.673 \times 10^{-27} \text{ kg}}$$

$$\mu_N = 5.05 \times 10^{-27} \text{ J C s kg}^{-1} = 5.05 \times 10^{-27} \text{ J T}^{-1}$$

3. The Larmor precessional frequency, $\nu = \frac{g_N \mu_N B_0}{h}$

Substituting the values of different parameters we get,

$$\begin{aligned} \nu &= 5.585 \times 5.05 \times 10^{-27} \text{ J T}^{-1} \times 7.1 \text{ T} / 6.626 \times 10^{-34} \text{ J s} \\ &= 3.02 \times 10^8 \text{ s}^{-1} \\ &= 302 \times 10^6 \text{ Hz} = 302 \text{ MHz} \end{aligned}$$

4. i) This molecule is expected to give two signals; one for the aromatic protons of the phenyl group and other for the methyl group.
- The phenyl protons signal is expected to be in the range of ~ 7.2 ppm and is likely to be a singlet. The methyl group protons are also expected to be a singlet and are expected to be downfield due to the effect of phenyl group. These may appear around $\delta = 2.3$ to 2.6 ppm.
- ii) Ethanoic acid is a simple molecule and is expected to give two signals, one for the methyl group and the other for the acidic proton of carboxylic acid. The methyl group protons are expected to appear as a singlet around 2 ppm whereas the acidic protons are likely to be highly downfield in the range of 10-12 ppm.
- iii) Methyl isopropyl ketone is slightly more complicated as compared to the above two molecules. The methyl protons are expected to be slightly downfield (~ 2 ppm) as compared to the alkyl protons and would appear as a singlet as these will not couple with other protons. The isopropyl part of the molecule would give two signals. One of these is expected to be a multiplet for one proton. As this proton is close to a carbonyl group, it is likely to appear downfield. The other signal would be for six protons and is expected to appear around 1-2 ppm as a doublet.
5. i) **Shielding constant:** The electrons circulating around the nucleus induce a magnetic field that opposes the applied field or shields the nucleus from the applied field. The shielding constant σ is a measure of the extent of shielding of the nucleus by the electrons around it.
- ii) **Anisotropy of chemical bond:** The chemical shift is dependent on the orientation of the NMR active nucleus with respect to the neighbouring bonds especially the π -bonds. The circular motion of π electrons in the presence of applied magnetic field generates induced magnetic field which opposes the applied field for certain parts of the molecule and for other parts it augments the applied field. This differential field strength of the induced field in different directions is called anisotropy of the bond.
- iii) **Larmor frequency:** When placed in an external field, the magnetic moment of the spinning nuclei interacts with the field and the resulting

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torque makes the nuclear magnetic moment vector to precess around the direction of the applied field. The frequency of this precession motion is called Larmor frequency.

- iv) **Coupling constant:** The interaction of neighbouring protons in the molecule causes the signals of the interacting protons to split. The magnitude of splitting is proportional to the strength of the interaction. The separation of these peaks in frequency units is called coupling constant and is independent of the strength of the applied magnetic field.

6. Structure of the compound is $\text{CH}_3\text{-CH}_2\text{-CH (Br)-COOH}$.
7. The lines due to chemical shift can be differentiated from those due to spin-spin splitting by taking the spectrum at a higher field. Though the chemical shift in terms of δ units does not change but in terms of shift from the internal standard in units of Hz would increase. On the other hand the coupling constant being independent of the applied field its separation would not get affected by changes in the field strength.

is NMR an Absorption or Emission technique ?

The measurement of a NMR spectrum signal is very different from infrared or UV-VIS spectroscopy in which absorption of photons is measured as a function of frequency. It has been argued that NMR spectroscopy is similar to emission spectroscopy, in that absorption of radiowave frequencies is detected indirectly by the subsequent emission of radiowaves at the same frequency. While this explanation can account for a simple proton NMR spectrum, it does not readily explain pulsed experiments that have given NMR its tremendous ability to uncover couplings and correlations among nuclei in a molecule. Furthermore, counterarguments have been made that the small NMR probe would be a very poor emitter and receiver of long wavelength radiofrequency radiation and the relatively low radiofrequency implies extremely low absorption and emission probabilities. Hence, it is better to view NMR spectroscopy as a magnetic induction experiment rather than as an experiment involving absorption and emission of photons.

A more complete understanding of NMR spectroscopy is achieved if one considers the effect of applied magnetic fields on the nuclear magnetic moments in the system. Briefly, an NMR sample in an external magnetic field is disturbed from equilibrium by application of a transverse secondary magnetic field, and its response to the disturbance is recorded. Furthermore, a tremendous enhancement of signal-to-noise, as well as the ability to carry out multiple-pulse experiments, occurs when a pulse of radiowave frequencies is used to disturb the system from equilibrium.

The time dependence of the return to equilibrium is then measured, from which frequency spectrum is generated via a Fourier transform.

In this section the underlying physical basis of the NMR phenomena is described. The implementation details in a modern spectrometer are covered in a subsequent section.